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Local Rearrangement of the Iodide Defect Structure Determines the Phase Segregation Effect in Mixed-Halide Perovskites

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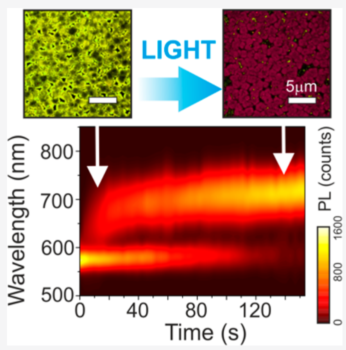
ABSTRACT: Mixed-halide perovskites represent a particularly relevant case within the family of lead-halide perovskites. Beyond their technological relevance for a variety of optoelectronic devices, photoinduced structural changes characteristic of this type of material lead to extreme photophysical changes that are currently the subject of intense debate. Herein we show that the conspicuous photoinduced phase segregation characteristic of these materials is primarily the result of the local and metastable rearrangement of the iodide sublattice. A local photophysical study comprising spectrally resolved laser scanning confocal microscopy is employed to find a correlation between the defect density and the dynamics of photoinduced changes, which extend far from the illuminated region. We observe that iodide-rich regions evolve much faster from highly defective regions. Also, by altering the material composition, we find evidence for the interplay between the iodide-related defect distribution and the intra- and interdomain migration dynamics giving rise to the complexity of this process.



With efficiency reports in solar cells[1] and light-emitting devices (LEDs)[2] pairing or outperforming any other solution process technology achieved over less than a decade, lead halide perovskites (LHPs) have emerged as one of the most technologically relevant materials of the 21st century. However, one of the main bottlenecks for their commercialization is their instability against external stimuli and, in particular, against illumination. In this respect, a material that epitomizes such instability is mixed-halide perovskites (mLHP's) with general formula $CH_3NH_3Pb(I_xBr_{1-x})_3$ whose electronic band gap can be tuned across the visible[3] by using a combination of halides in the appropriate ratio to form solid alloys that can be incorporated into tandem solar cells,[4] LEDs,[5] or lasers.[6] Upon illumination, mLHP's undergo structural changes leading to the formation of I-rich regions with composition $CH_3NH_3Pb(I_{0.8}Br_{0.2})_3$[7] (a further compositional change being prevented by the presence of a phase transition[3]) toward which photogenerated carriers are funneled, negatively affecting the performance of photovoltaic devices and LEDs.[8−10]

While the ultimate mechanism behind the formation of I-rich domains is still under debate, some factors have been identified to play a key role. The observed spectral changes involve a redistribution of the halide component through ionic migration, and therefore the presence of halide-related defects is necessary. Likewise modifying the defect density can influence the magnitude and dynamics of the process,[11−14] and most routes to prevent these photoinduced changes[4,5,15−21] are associated with the blocking of migration routes. The microstructure of the material also determines the dynamics of the process, where the presence of grain boundaries (GB) has been shown to promote the photophysical changes,[5,22−24] and thus large crystals tend to be more stable. This matches reports of the formation of I-rich domains at GBs,[23,25,26] although it has been shown that they can also appear throughout the bulk.[27,28] On the other hand, when the dimension of the material is reduced down to the nanometer scale, photophysical changes are largely suppressed.[29−31]

The general consensus is that mLHP films contain an initial distribution of entropically mixed halides. Due to the presence of a miscibility gap at room temperature,[32] external stimuli provide the driving force for halide ions to move from their original position leading to the formation of I-rich domains. Several proposals have been presented as the driving force for such structural changes, including the electrostatic interaction between charges trapped at defects and spontaneously formed I-rich domains,[20] the presence of positive[22] or negative[21] charges at grain boundaries, a photoinduced strain gradient created by a uneven charge carrier generation pattern,[11] and local strain induced by the formation of large polarons.[26]

In this work, we present a photophysical study of mLHP where the dynamics of the spectral changes are extracted locally and correlated with the defect density. We find that the formation of I-rich domains extends well beyond the photoirradiated regions and follows a complex behavior where structural changes within single-crystalline grains and

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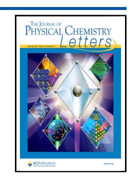

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across grains have to be taken into account. Finally, the sample stoichiometry is modified by varying the halide-to-metal ratio in order to elucidate the nature of the defects behind the process, finding evidence that supports the fact that iodide-based defects are the ones playing a key role in the process. The present study highlights the need for local probes in order to correlate the many factors influencing the instability of mLHP. Furthermore, it unveils the complexity of this process, which represents an extreme case of one of the main issues for LHP commercialization, consisting of photoinduced changes under external illumination.

The study of photoinduced changes in mLHP was carried out on $CH_3NH_3Pb(I_{0.17}Br_{0.83})_3$ films. Details of sample fabrication can be found in section S1.1 of the Supporting Information (SI). In an initial step, we measured the PL from the samples using a combination of a CW (λ = 450 nm, 0.01 W/cm^2) and a pulsed (λ = 450 nm, 0.01 W/cm^2, 100 fs pulses, 1 kHz repetition rate) laser (section S1.2 of the SI). In a typical experiment, the sample is initially illuminated with the pulsed laser, and only the PL peak associated with the mixed phase, peak 1, is observed (stage I in Figure 1), evidencing the

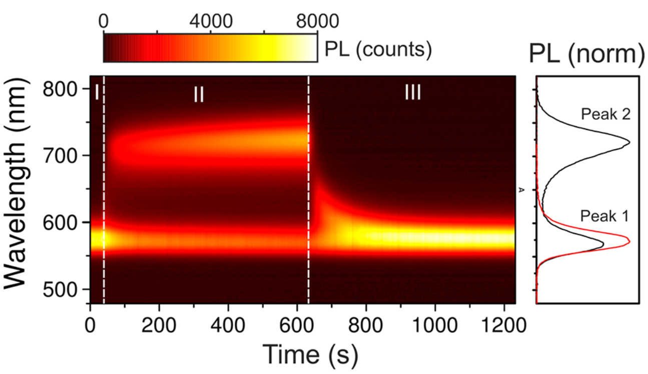

Figure 1. Color plot showing the time evolution of the PL from a $CH_3NH_3Pb(I_{0.17}Br_{0.83})_3$ film when illuminated with a pulsed kHz laser (stage I), the combination of a pulsed and CW laser (stage II), and finally the pulsed laser again (stage III). The right panel shows normalized PL spectra at the end of stages I (red) and II (black curve), where peaks 1 and 2 from mixed (I-rich) phases are highlighted.

absence of photoinduced changes as previously observed with low-repetition-rate sources.[33] Next, the CW beam is introduced, and a reduction in the intensity of the initial PL peak is accompanied by the emergence of a red-shifted component, peak 2, associated with the formation of I-rich regions (stage II) with a composition of $Br_{0.2}I_{0.8}$ (section S1.3 and Figure S1 in the SI). Upon removal of the CW beam, the sample returns to its original optical response over a longer period (stage III).

Control measurements were carried out to evidence that the pulsed laser does not introduce any photoinduced change (section S2, Figures S2 and S3 in the SI) and serves to give qualitative information on the composition of the film during the light-induced changes (Figure S4). Furthermore, small (below 3 nm) spectral changes in peak 1 during stage II in Figure 1 evidence that the composition of the majority mixed-phase barely changes, pointing to the absence of a phase in which the proportion of Br is larger than the stoichiometric 83% expected. (See the discussion in section S2 in the SI and Figure S4.) As the amount of I-rich material formed upon irradiation has been estimated to be in the 5−20% vol. range,[7,29,34] the lack of a Br-enriched component indicates that photoinduced changes are actually triggered by a local rearrangement of iodide atoms where iodide vacancies and interstitials recombine, leading to an I-rich lattice.

Next we performed a local PL study using a laser scanning confocal microscope (LSCM) with spectral detection. (See Section S1.4 in the SI for details.) Figure 2 shows true-color PL

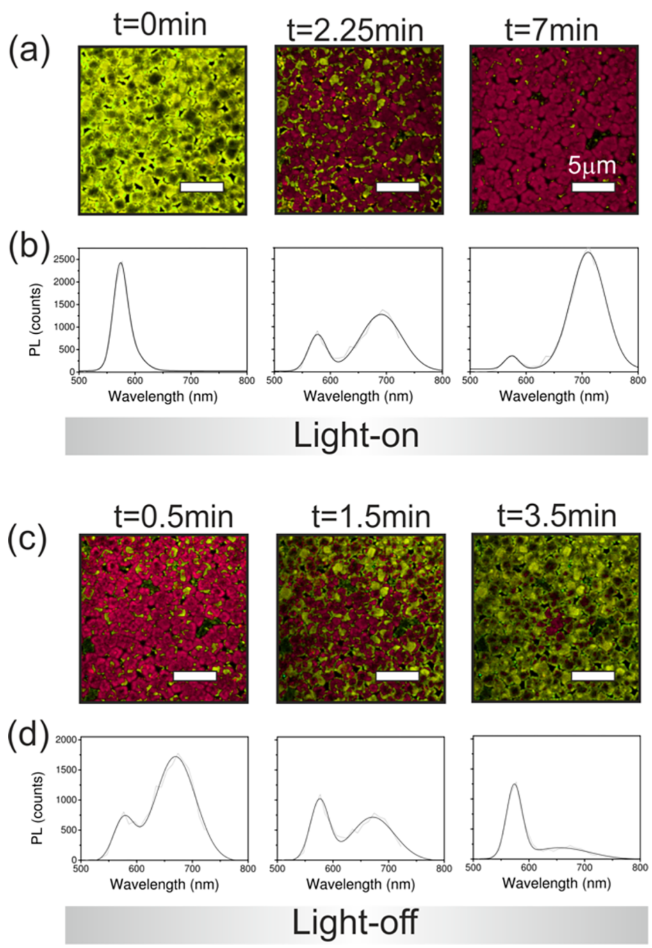

Figure 2. True-color PL images and spectra for an mLHP film tracking its evolution during (a, b) irradiation and (c, d) recovery. Scale bars in all PL images correspond to 5 μm. Gray (black) curves correspond to experimental data (fit to two Gaussians).

images, each corresponding to a line scan of the region under study, together with PL spectra at different times showing an evolution similar to that in the macroscopic measurements of Figure 1. The initial image of the sample (t = 0 s) presents an inhomogeneous distribution of the PL intensity, which can be associated with a distribution of crystalline defects introducing intragap states responsible for nonradiative recombination paths for the photogenerated carriers.[35] PL spectra collected at different points of the image show a similar spectral position, indicating an identical Br/I ratio across the sample. (See Section S3 and Figure S5 in the SI.) The spectra in Figure 2b,d were fitted to a double Gaussian, and the time evolution of peak 2 was retrieved to illustrate the dynamics of the process of formation of I-rich regions. The time evolution of $I_{peak\,2}$ could be fitted to single-exponential growth/decay from which a rate

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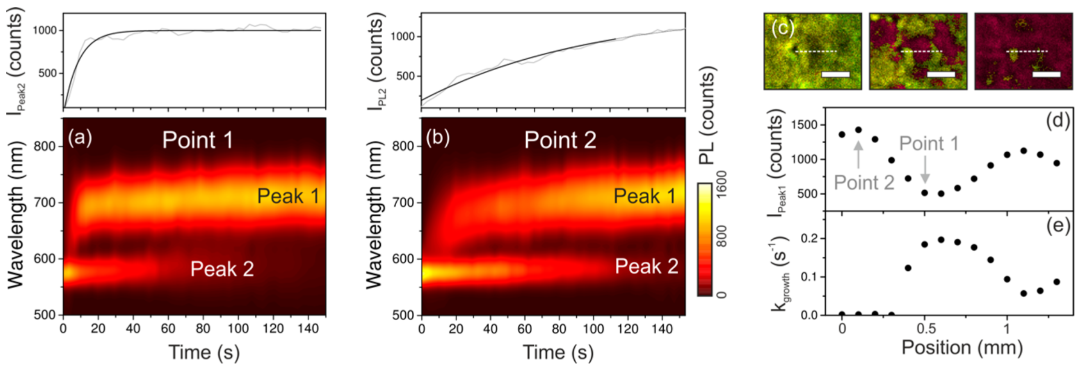


Figure 3. PL as a function of time from a small domain in a $CH_3NH_3Pb(I_{0.17}Br_{0.83})_3$ film. (a) Color plot showing the time evolution of the PL from a dark region where $I_{peak\,2}$ is shown in the top panel. (b) Same data for a bright region. (c) True color PL images of the domain under consideration for three illumination times (scale bar is 1 μm). (d) and (e) show initial $I_{peak\,1}$ from the mixed phase and growth rate of I-rich $I_{peak\,2}$, respectively, across the dashed line in (c). Points 1 and 2 correspond to those from which spectral information is shown in (a) and (b).

k_{growth} ($k_{recovery}$) of 0.017 (0.011 s^{-1}) is obtained (Section S4 and Figure S6).

From the PL images in Figure 2a, it becomes clear that regions with a low initial PL intensity (i.e., high defect density) undergo a faster transition to a PL dominated by I-rich domains. To further establish this correlation, we extracted the dynamics of the photoinduced changes across a domain, comprising bright and dark regions (Figure 3). Plotting the time-evolution of the spectral response for a dark (Figure 3a) and a bright (Figure 3b) spot, we see how the transition to a PL dominated by I-rich regions is faster in the former. To better appreciate this correlation, the spatial variation of the optical response across a domain (Figure 3c) was obtained. The peak 1 intensity (Figure 3d) was extracted and compared with the activation constant, k_{growth} (Figure 3e), retrieved from the time evolution of peak 2 at each location. The spatial variation of the PL from the mixed phase is clearly the inverse of that of the dynamics of the photoinduced process.

Establishing a similar trend for the recovery is not straightforward. While initially dark regions show a slower recovery, now the PL from the area under consideration (Figure 2c) shows an additional spatial inhomogeneity with material recovery taking place at a faster pace for regions far from the image center. The combination of both dynamics determines the sample spectral behavior during the recovery. This radial distribution is not associated with the illumination pattern, which takes place through a sequential line scanning confined within the measurement area shown in Figure 2a,c. It likely points to structural changes extending beyond the illuminated area. This was confirmed by performing an irradiation cycle similar to that of Figure 2a and then opening the field of view of the microscope and collecting a series of PL images (Figure 4). Here it can be clearly seen how the formation of I-rich domains associated with the migration of halide ions extends beyond the illuminated area (section S5 and Figure S7 in the Supporting Information), pointing to a long-range driving force as will be discussed below.

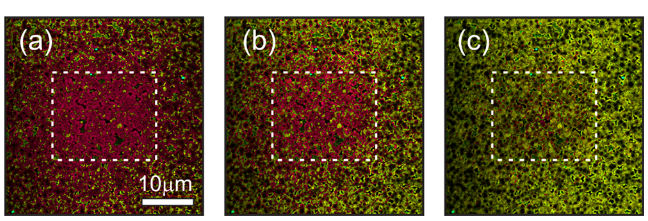


Figure 4. Expanded confocal PL images taken after irradiating the region highlighted by the dashed line. Images were taken after 0.5 (a), 1 (b), and 3 (c) minutes.

To further explore the nature of the defects behind these changes, we fabricated a set of samples where the composition was changed by keeping the Br/I mol ratio constant but varying the halide-to-metal ratio $R_H = (n_{Br} + n_I)/n_{Pb}$ above and below the stoichiometric value $R_H = 3$ in order to modify the density of halide-related defects. Details of the film fabrication are given in section S1 of the SI. Energy-dispersive spectroscopy (EDS) measurements show that the samples have a slightly lower ratio (R_H*) than the one contained in the precursors (Figure S8).

Next, the PL from these samples was collected following illumination/recovery cycles such as the ones shown in Figure 2 over 20×20 μm^2 regions. Figure 5a shows the initial peak 1 intensity (corresponding to the mixed phase) of the samples after the first laser scan for different values of R_H*, showing a monotonic increase pointing to a reduction in the density of defects introducing deep-trap states, which have been associated with halide interstitials and lead vacancies in the bulk[36] as well as at GB.[37] In principle, one would expect that increasing R_H should increase the density of halide interstitials. To solve this apparent contradiction, we next consider the spectral response of peak 1 (Figure 5b) and observe how increasing R_H* also leads to a slight red shift pointing to a larger fraction of iodide incorporating into lattice positions. This trend was also observed when studying the synthesis of the films of different stoichiometry with XRD (Section S6.2 and Figures S9 and S10 in the SI). This evidence indicates that as R_H increases, the total amount of halide in the sample increases but in a different manner depending on its nature. Iodide is incorporated into lattice positions, as indicated by PL and XRD results. On the other hand, additional bromide would be incorporated either in interstitials or as part of an amorphous phase.

Next, the growth/recovery rates of peak 2 were extracted for samples with different R_H* values (Figure 5c). We see how k_{growth} decreases with R_H*, which according to the above discussion implies that iodide defects most influence the photoinduced changes in mLHP films. This can be explained as vacancies have been demonstrated to be the main route for halide migration in LHP[38] and also for the formation of I-rich

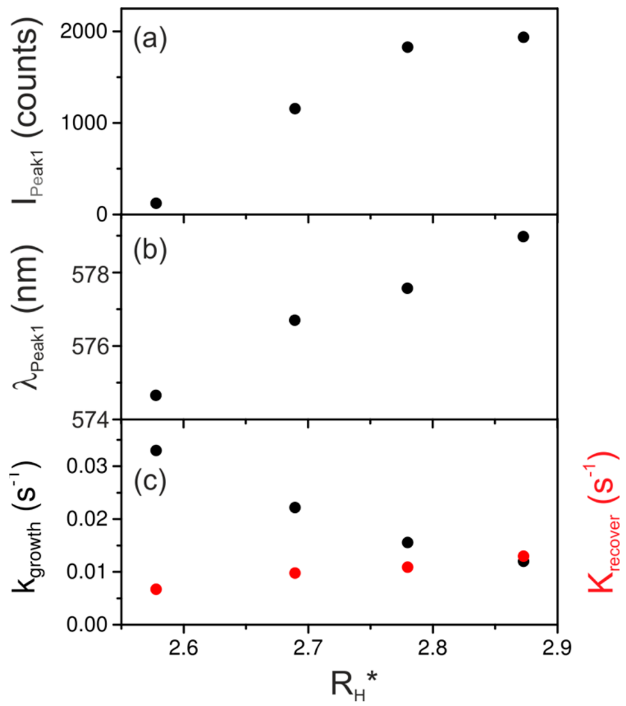


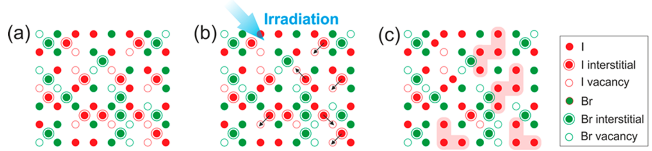
Figure 5. (a) Evolution of the initial peak 1 intensity and (b) spectral position of an mLHP film and (c) the rate of growth/recovery of peak 2 with respect to the halide-to-metal ratio R_H* estimated from EDS.

domains in mLHP,[14] albeit halide migration through interstitials has also been shown to be relevant for LHP.[39−42]

Finally, the recovery rate $k_{recover}$ of the optical response of the different films shows an inverse trend to that of k_{growth}, increasing as R_H* grows (Figure 5c). Such recovery, as shown in Figure 2c, takes place following a completely different spatial pattern with a radial distribution of the spectral changes pointing to the movement of ions across domains. Upon observation under scanning electron microscopy (SEM), it can be appreciated (Figure S11) that decreasing R_H influences the sample morphology, where the connectivity between domains is increased as the halide-to-metal ratio increases. Assuming interdomain migration to take place through GBs agrees with recent reports of these extended defects behaving as efficient routes for ion migration.[43,44]

According to the results presented herein, the driving force behind the formation of I-rich domains needs to have a long-range spatial extent in a direction normal to that of the incident pump beam. This matches the diffusion of photogenerated charges, either as free carriers or as large polarons,[45] which have been reported to span values from a few micrometers to above 100.[46−50] Furthermore, the presence of defects promotes the formation of I-rich regions, which matches the mechanism of carrier trapping proposed in ref 20, but also could favor the formation of large polarons.[51] In particular, we have seen that it is iodide-related defects that would have a combined role of trapping charges and providing the main migrating species, matching recent reports pointing at this element as the most mobile species under illumination in mLHP.[52] Thus, under light illumination the crystalline lattice would be partially repaired mainly by the formation of regions where iodine vacancies/interstitials recombine, leading to the formation of I-rich regions (Scheme 1). This would agree with the absence of a Br-rich phase observed by us and others. Nevertheless, the presence of Br-rich regions reported in the literature is also feasible if the synthesis leads to a majority of initial Br-related defects. The relevance of morphology and defect structure is likely related to the recent observation of intermediate I-rich regions.[53]

Scheme 1. Proposed Mechanism Responsible for Photoinduced Phase Segregation[a]



[a]An initial distribution of halide interstitials and vacancies is modified under external illumination leading mainly to the recombination of I-related defects ending in the formation of I-rich regions.

Finally, upon removal of the external illumination, the system returns to its original configuration driven by entropy. At a local level, initially defective (dark) regions that underwent pronounced photoinduced changes take longer to return (Figure 2b). On a macroscopic scale, the recovery dynamics are dominated by a radial pattern, evidencing an intergrain ionic migration strongly linked to the sample domain connectivity, likely taking place through GBs where enhanced ion migration has been reported.[43,44] This shows that, while most photophysical experiments on mLHP are performed on a macroscopic scale, a correct interpretation requires taking into account the sample morphology as it strongly determines phase segregation through an interplay of bulk and GB processes.

In summary, a local photophysical study of mLHP films has been carried out where a direct correlation between the defect density and I-rich region formation has been established. Employing local photophysical characterization on samples with different stoichiometry, we have found evidence that iodide-related defects play a key role in the process. Furthermore, the extent of the observed changes, which occur far beyond the illuminated region, point to photogenerated charge carriers, either as free carriers or in the shape of large polarons, which would be responsible for the driving force as they are trapped by defects. Finally, the recovery dynamics point to a complex scenario where combined inter- and intradomain processes, strongly dependent on sample morphology, are taking place. These results explain the origin of the process that gives rise to the conspicuously observed phase segregation of mixed halide perovskites and reveal that a precise knowledge of the sample morphology and defect structure is needed to disentangle the contribution from different atomic components and their migration routes. The relevance of the present work further relies on the fact that phase segregation in mLHP is part of a broader issue comprising the photoinduced halide migration whose mitigation is fundamental for future applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpclett.0c01127.

Experimental (film fabrication, PL spectroscopy experiments, estimation of composition from I-rich domains, and LSCM experiments); PL spectra from mLHP films; spectral homogeneity in confocal images; growth/

recovery dynamics; spectral information from the region surrounding the illuminated area; and films with different stoichiometry (EDS studies, film formation, and SEM) (PDF)

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Notes

The authors declare no competing financial interest.

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